Cross-Environment Behavioral Fingerprinting for IoT Device Security

A Cautionary Tale in Evaluation Methodology

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Why Cross-Environment Transfer Matters

Most IoT intrusion detection research trains and tests on one network. This project asks a more basic question:

> *Does a behavioral fingerprint learned on one IoT network actually detect threats on a completely different one?*

Two homes

rarely behave alike on the wire

Two datasets

IoT-23 and CICIoT2023, independently collected

One question

what actually transfers, and what doesn't

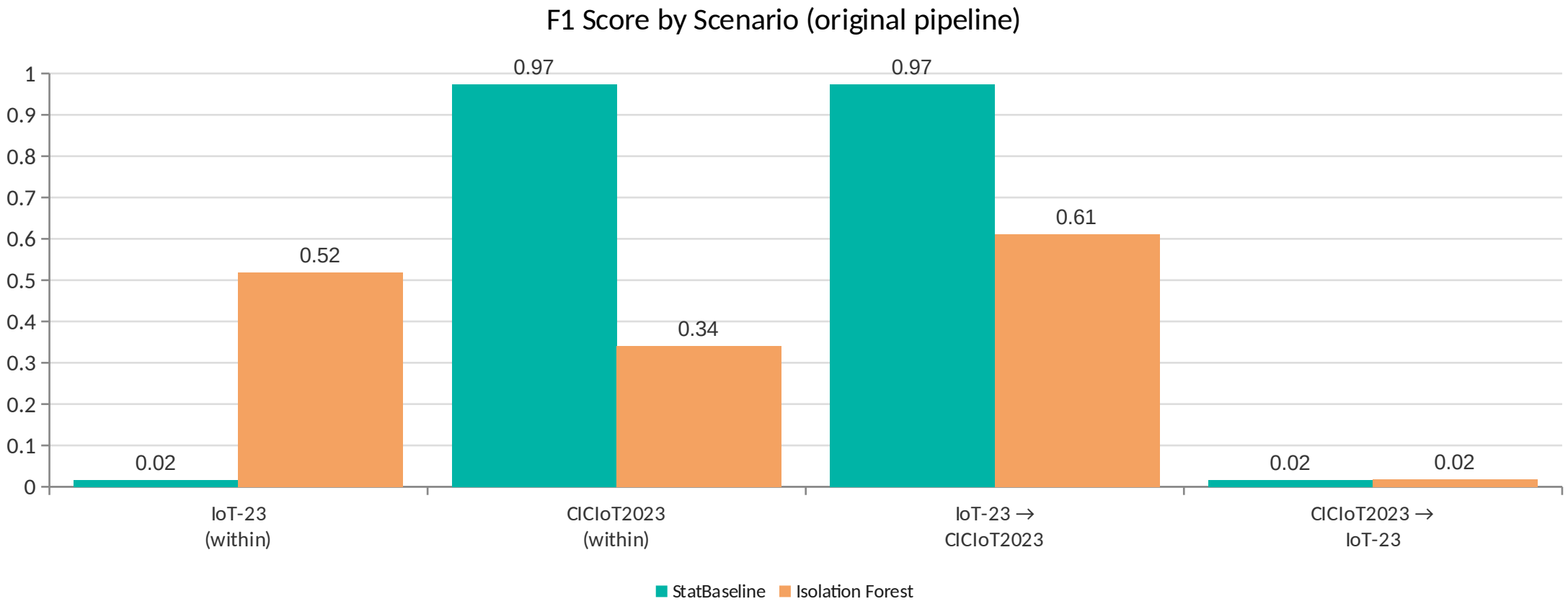
Datasets and Approach

- **IoT-23** — 5 scenarios: 2 benign honeypot captures, 3 malware captures
- **CICIoT2023** — 8 merged CSV files, 33 attack types across real IoT devices
- **Shared feature schema** — duration, total bytes, total packets, avg packet size, protocol flags
- **Three models** — a statistical z-score baseline, K-Means clustering, and an Isolation Forest
- **Training** — all three models trained exclusively on benign traffic

Four Early Fixes Before We Could Trust Any Number

1. **Schema mismatch** — IoT-23 header parsing broke on mixed tabs/spaces; CICIoT2023 was missing a duration field entirely
2. **Class imbalance** — CICIoT2023 test data was 97.7% attacks — inflated every score to near-perfect
3. **Feature-scale mismatch** — duration proxies differed by 7 orders of magnitude across datasets → total false-positive collapse
4. **Categorical instability** — a zero-variance protocol flag produced infinite z-scores on unrelated traffic

Initial Results: It Looked Like Transfer Worked



StatBaseline appeared to transfer genuinely well. The next three slides explain why that was wrong.

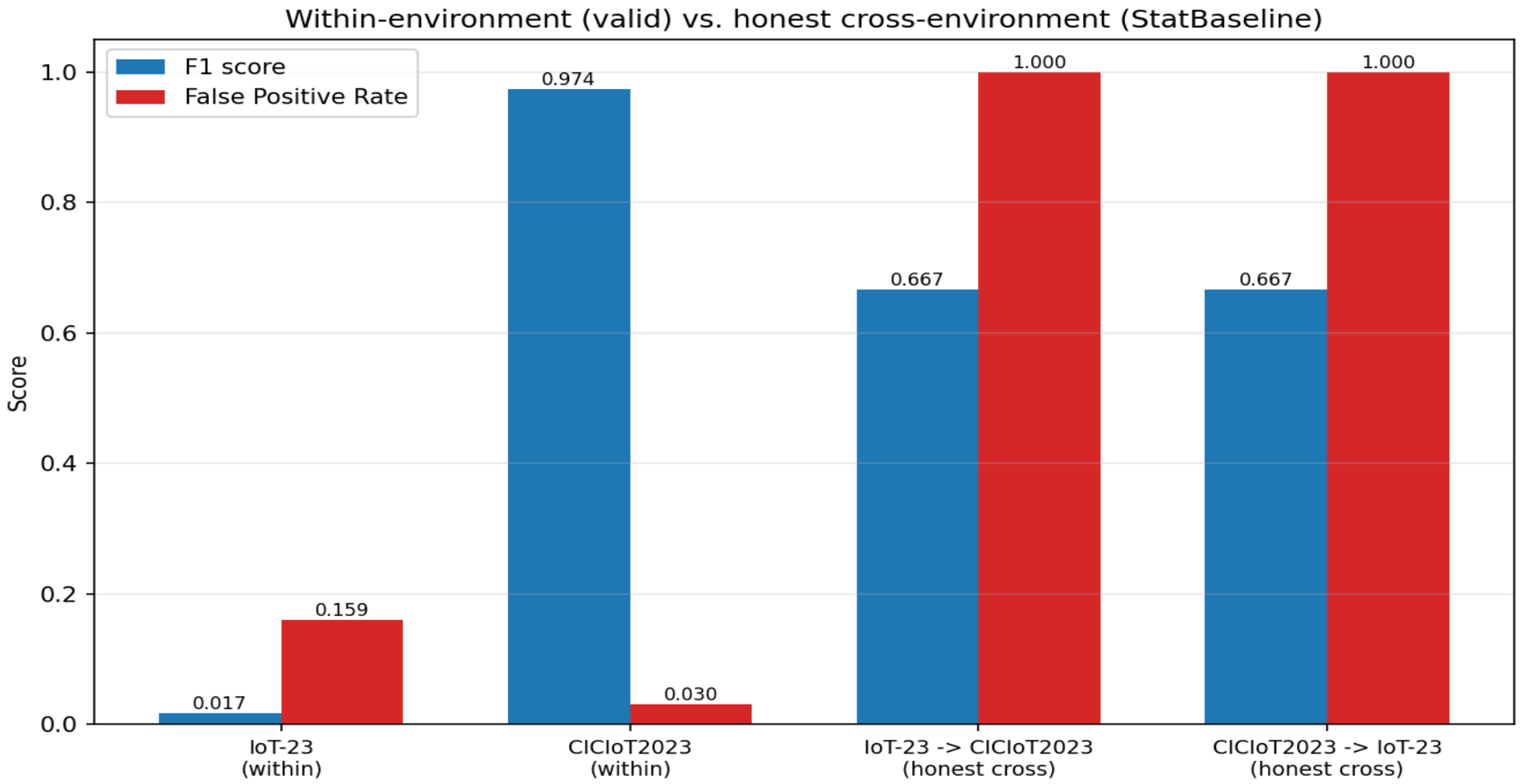
Wait — Something Was Off

A suspicious pattern: within-environment and cross-environment F1 for the SAME test dataset were nearly identical, no matter which dataset supposedly trained the model.

Within IoT-23 **0.017**	≈	Cross CICIoT2023 → IoT-23 **0.017**
Within CICIoT2023 **0.974**	≈	Cross IoT-23 → CICIoT2023 **0.974**

We verified directly: standardizing training data against its own mean/std before fitting makes StatBaseline's learned parameters ALWAYS mean=0, std=1 — regardless of which dataset trained it.

Honest Re-Evaluation: The Real Result



Root Cause: Two Datasets, Two Different Measurements

CICIoT2023 benign total_pkts

std = 0.125
99.95% of values exactly 10

IoT-23 benign total_pkts

std = 526
median = 1, max = 363,141

> *“We group the values captured in window sizes of 10 and 100 packets... and calculate their mean values.”*
>
> — Neto et al. 2023, CICIoT2023's own methodology paper

CICIoT2023's “packet count” is a fixed aggregation window, not a measurement of flow length. No standardization can make two different measurements comparable.

Not All Models Fail the Same Way

We verified directly: fit two instances of each model (one per dataset), apply both to identical test points, measure agreement.

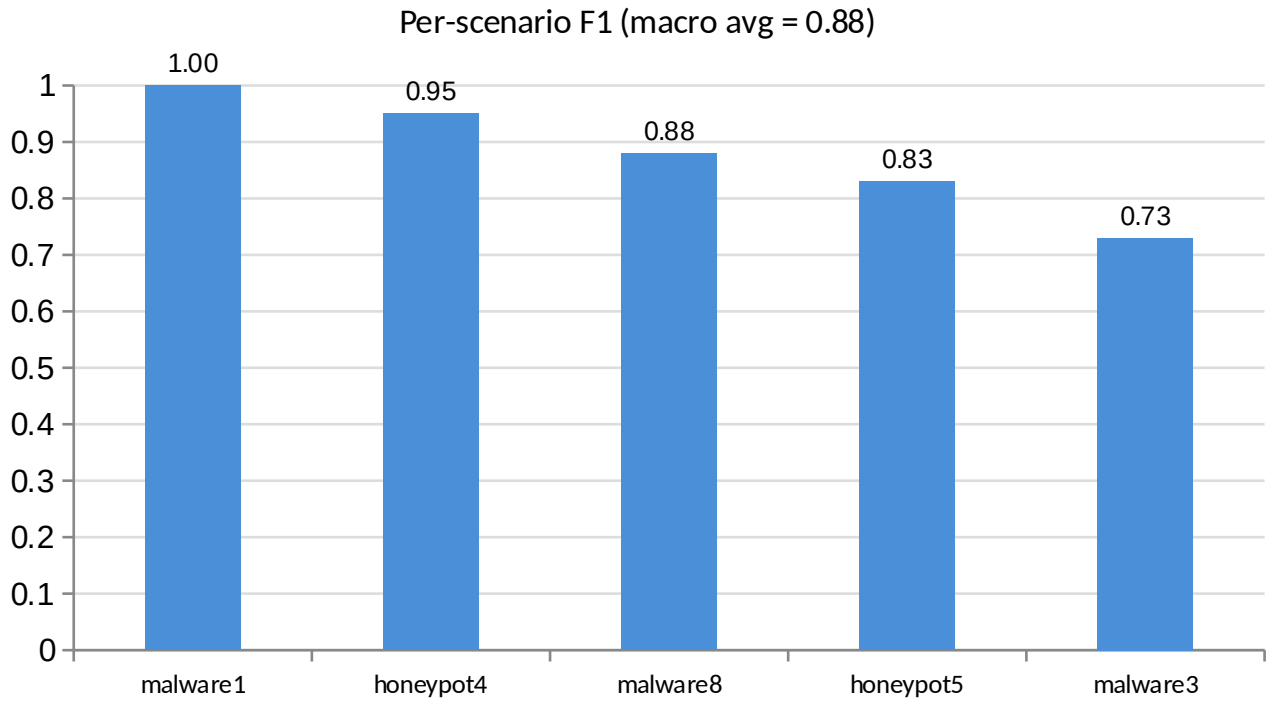
Model	Agreement Rate	Interpretation
StatBaseline	~1.000 (by construction)	**Fully degenerate**
Isolation Forest	0.356	**Genuine**
ClusterBaseline (K-Means)	0.948	**Substantially degenerate**

Isolation Forest's cross-environment failure is REAL and independently verified. ClusterBaseline's is mostly the same artifact that fooled StatBaseline.

Low agreement = the two training sets taught the model genuinely different things. High agreement = training source barely mattered.

A Complementary Finding: Scenarios ARE Fingerprintable

Sidestepping the cross-dataset problem entirely: can we tell IoT-23's 5 capture scenarios apart using only their own benign behavior?



What distinguishes scenarios?

duration: 59.7%

total_pkts: 17.4%

avg_pkt_size: 9.4%

total_bytes: 9.0%

protocol flags: <2%

Timing dominates — not protocol choice

Once evaluated honestly:

No model achieves genuine cross-environment transfer

Root cause: the two datasets measure “packet count” differently — not just at a different scale.

But this evaluation artifact affects models unevenly: fully explains StatBaseline, substantially (not fully) explains K-Means, and does not explain Isolation Forest's independently-verified failure.

Within-dataset scenario fingerprints, by contrast, are real (macro F1 = 0.88).

Limitations and Future Work

Limitations

- IAT-as-duration-proxy turned out to be a deeper problem than scale alone
- Scenario fingerprinting stays within one dataset — doesn't prove cross-dataset device ID
- Model hyperparameters fixed, not tuned
- No autoencoder comparison attempted

Future Work

- Audit other feature pairs for the same construction-incompatibility risk
- Test whether a 3rd, structurally different dataset confirms this pattern
- Investigate why K-Means is partially (not fully) degenerate under self-calibration
- Verify scenario labels against IoT-23's own device documentation

Thank You

Questions?